Let a = the sum of last three numbers in your Student ID, write a matrix that has trace a

Find its eig vectors and eig values

$a = 8$

$$\begin{bmatrix} 1 & 1 & 4 \\ 0 & 3 & 5 \\ 0 & 0 & 4 \end{bmatrix} = A$$

$$(2-\lambda)\ \det(A - \lambda I) = 0$$

$$\det\begin{vmatrix} 1-\lambda & 1 & 4 \\ 0 & 3-\lambda & 5 \\ 0 & 0 & 4-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)(3-\lambda)(4-\lambda) = 0$$

$$\lambda = 1, 4, 3$$

$\lambda = 1$

$$\begin{bmatrix} 0 & 1 & 4 \\ 0 & 2 & 5 \\ 0 & 0 & 1 \end{bmatrix} x = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$\lambda = 3$

$$\begin{bmatrix} -2 & 1 & 4 \\ 0 & 0 & 5 \\ 0 & 0 & 1 \end{bmatrix} x = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

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